

Discontinuous Galerkin method applied to critical phenomena in gravitational collapse

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Numerical studies of critical phenomena in gravitational collapse have provided key insights into the nonlinear dynamics of Einstein’s field equations in the strong-field regime. However, most prior investigations have been restricted to highly symmetric models, limiting our understanding of the universality of critical behavior. Extending these studies to more general systems poses substantial computational challenges, as resolving the fine-scale structure near the black hole threshold requires extremely high spatial and temporal resolution. In this work, we present the first direct comparison between the discontinuous Galerkin (DG) and finite-difference methods applied to critical phenomena. For the case study, we used the classical example of spherically symmetric gravitational collapse of massless scalar fields. The DG method, which combines the high-order accuracy of spectral techniques with the flexibility and locality of finite-difference schemes, demonstrates significant computational advantages: Near criticality, it achieves comparable accuracy with fewer degrees of freedom, approximately half the number of computational steps, and a third of the run-time required by finite-difference methods. These improvements mitigate the run-time and resolution limitations characteristic of finite-difference simulations in higher dimensions and complex field systems. A detailed extension of the DG framework to the collapse of nonlinear, axisymmetric gravitational waves will be reported in future work.

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I. INTRODUCTION

General relativity has been remarkably successful in precisely describing gravitational effects on all observed scales. These include phenomena within the Solar System [1], binary pulsar timing [2,3], gravitational wave observations [4] and large-scale cosmological structure [5].

Nevertheless, some predictions associated with the extreme spacetimes resulting in the formation of black holes call into question the viability of general relativity as a final theory. For example, the inevitability of singularities [6,7] indicates a breakdown of classical spacetime and suggests the need for a quantum theory of gravity [8,9]. Penrose [10] proposed the cosmic censorship conjecture, according to which these singularities are always contained within a black hole’s event horizon and thus can have no observable consequences. While this conjecture appears to be at least *nearly* true, evidence from theoretical and numerical studies [8,11–15] points to the possibility of naked singularity formation under specific nongeneric initial conditions.

Indeed, given the infeasibility of laboratory-based experimentation on black holes, numerical “thought experiments” currently provide the best testing ground for the most extreme predictions of general relativity.

In 1993, Choptuik’s seminal paper on critical phenomena in gravitational collapse of a massless scalar field in spherical symmetry [12] identified an infinitesimal point in phase space in which this simple system could yield a naked singularity. Near this point, black holes can be created with a mass that scales as

$$M_{\text{BH}} \simeq c_f |p - p^*|^\gamma, \quad (1)$$

where c_f is a family-dependent constant, p is a parameter of the initial data that defines the phase space and p^* is the infinitesimal point that separates black hole formation and dispersion. The critical exponent γ is universal with respect to the initial configuration of the field and was originally measured to be $\gamma \approx 0.37$.

In the region of strong-field interaction, the solution also begins to exhibit discrete self-similarity which is again universal. Choptuik originally referred to this phenomenon as *echoing*. To match the exponential scale of self-similarity, Choptuik used similarity-adapted coordinates

$$\rho = \ln(r), \quad (2)$$

$$\tau = \ln(T_0^* - T_0), \quad (3)$$

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where T_0 is the central observer proper time and T_0^* is the accumulation time of the field parametrized by p^* , the perfectly critical value, as experienced by a central observer, and is specific to each family. The discrete spacing Δ between successive self-similar moments, or echoes, was measured to be $\Delta \approx 3.44$.

The interplay of mass scaling and discrete self-similarity led both Hod and Piran [16] and Gundlach [17] to independently predict and numerically verify the existence of fine structure in mass scaling beyond the critical exponent. The mass of resultant black holes is more fully described by the relation

$$\ln(M_{\text{BH}}) = \gamma \ln |p - p^*| + \Xi[\ln |p - p^*|] + c_f, \quad (4)$$

where Ξ is a periodic function with a universal period $\varpi = \Delta/(2\gamma) \approx 4.6$.

Shortly after Choptuik's paper, in an effort to study critical phenomena in more generic systems, Abraham and Evans [18] demonstrated similar behavior in the vacuum collapse of axisymmetric linearized gravitational waves. Since then, the goal of understanding fully nonlinear axisymmetric vacuum collapse (Brill [19] wave collapse) has been pursued, in broad terms, using both finite-difference (FD) [20–23] and spectral methods [23–25]. In the current paper, we take a first step in exploring the use of discontinuous Galerkin (DG) methods [26] in this endeavor.

Discontinuous Galerkin acts as a hybrid between finite-difference and spectral methods [27]. FD has, as an advantage, the locality in its definition of solutions and operators, allowing local refinement in areas with sharp features. Spectral methods, on the other hand, are global in the definition of operators and solutions, which, in general, allows for higher accuracy with fewer degrees of freedom. Degrees of freedom, in this paper, represent the total number of nodes that have an associated value. For FD, this accounts for both normal and ghost nodes as ghost nodes are needed for the adaptive mesh. DG represents a middle ground between FD and spectral methods by having discrete elements that allow for local definitions and refinements. However, it maintains flux terms between the elements and definitions of solutions and operators that hold across entire elements to regain some of the advantageous global properties characteristic of spectral methods.

To date, the DG method has not been widely used in published studies specifically targeting gravitational critical phenomena. Since Choptuik's work, most studies of critical phenomena have relied on FD methods using the adaptive mesh refinement technique for resolving features near the critical point or double-null coordinates. In addition, computational studies have usually been performed in spherical symmetry, or at most axisymmetry, for reasons of computational tractability. However, DG has been used in related areas, including general relativistic hydrodynamics [28–31], e.g. in the modeling of neutron star

mergers. The high-order accuracy [32] of the DG method could be beneficial for resolving fine-scale structure in critical phenomena.

A Galerkin approach has previously been applied to the scalar field system [33–36] where all studies employ a Galerkin collocation formulation (and notably lack adaptive mesh refinement). Galerkin collocation formulation differs from discontinuous Galerkin in that the residual is forced to vanish only at a discrete set of collocation points, continuity is enforced over the entire domain, and partial differential equations are solved in the strong form. It should also be noted that it is harder to demonstrate stability in the Galerkin collocation method than in the discontinuous Galerkin method in general. The Galerkin collocation method, similar to spectral methods [37], necessitates the introduction of multiple subdomains to achieve local refinement. However, unlike discontinuous Galerkin, communication between these subdomains is handled via complex transmission conditions in the mass matrix which must be strongly enforced when computing the spectral representation of each field. This significantly hampers parallelism and requires special treatment to avoid Gibbs phenomena near shocks or discontinuities. Although such spectral methods can achieve higher convergences with relatively fewer degrees of freedom in the case of smooth solutions, these advantages come at the cost of the aforementioned limitations.

We anticipate several challenges in implementing the discontinuous Galerkin method in critical phenomena studies, which we intend to address in this and future papers. DG is built for equations in conservative form and has difficulty with second-order operators, and there is currently a lack of robust tools suitable for DG in numerical relativity. In this paper, we choose Choptuik's original problem with a massless scalar field minimally coupled to gravity as a simple first case. This choice removes the problem of second-order operators, which we will address in future papers. In Sec. II, we present a cursory solution to equations not in conservative form; however, more complicated situations are not seen in this simple model. Overall, this paper serves as an exploratory analysis of the DG method's usefulness, specifically in critical phenomena of gravitational collapse, and as a stepping stone to a fully DG simulation of Brill wave collapse.

II. METHODS

A. The massless scalar field in curved spacetime

The minimally coupled massless scalar field is assumed to be non-self-interacting and satisfies the massless Klein-Gordon equation:

$$\nabla_a \nabla^a \Phi = 0, \quad (5)$$

where ∇_a represents covariant differentiation.

Following Choptuik [12], we adopt a metric of the form

$$ds^2 = -\alpha^2(r, t)dt^2 + a^2(r, t)dr^2 + r^2d\Omega^2. \quad (6)$$

Defining the auxiliary variables

$$\Psi = \frac{\partial\Phi}{\partial r}, \quad (7)$$

$$\Pi = \frac{a}{\alpha} \frac{\partial\Phi}{\partial t}, \quad (8)$$

we arrive at a set of evolution equations for the derivatives of Φ :

$$\frac{\partial\Pi}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{r^2\alpha}{a} \Psi \right), \quad (9)$$

$$\frac{\partial\Psi}{\partial t} = \frac{\partial}{\partial r} \left(\frac{\alpha}{a} \Pi \right). \quad (10)$$

Additionally, Einstein's equations with stress-energy tensor

$$T_{ab} = \nabla_a \Phi \nabla_b \Phi - \frac{1}{2} g_{ab} \nabla_c \Phi \nabla^c \Phi \quad (11)$$

result in a set of constraints on the dynamical variables a and α , namely,

$$\frac{1}{a} \frac{\partial a}{\partial r} + \frac{a^2 - 1}{2r} - 2\pi r (\Pi^2 + \Psi^2) = 0, \quad (12)$$

$$\frac{1}{\alpha} \frac{\partial \alpha}{\partial r} - \frac{1}{a} \frac{\partial a}{\partial r} - \frac{a^2 - 1}{r} = 0, \quad (13)$$

$$\frac{\partial a}{\partial t} = 4\pi r \alpha \Psi \Pi. \quad (14)$$

B. Numerical solutions of the scalar field equations

For the studies presented in this paper, two independent codes were developed using the discontinuous Galerkin and Finite-Difference methods, respectively. These codes are described in the subsections below. As a cross-check, we also made use of an independent code developed by this group for previous publications [38,39].

1. Nondynamic equations

Analysis begins by selection of an initial field $\Phi(r, t = 0)$ from Table I. In each case, the amplitude A is the parameter that is varied to study critical behavior. Time symmetry is assumed, that is, $\Pi(r, t = 0) = 0$.

With this selection, initial data for the dynamical variables a and α are determined using Eqs. (12) and (13). This calculation is performed essentially the same

TABLE I. The initial conditions used for three different families for the scalar field Φ .

Family	Initial form of Φ
1	$\Phi(r) = A \exp\{-(r/5.35)^2\}$
2	$\Phi(r) = A \exp\{-(r-3)^2\}$
3	$\Phi(r) = A \tanh(r-3)$

way for both the DG and FD codes, using the fourth-order Runge-Kutta [40,41] method.

As a final condition to fix the coordinates, a is set to unity at the origin. This has the effect of enforcing regularity at the origin. Equation (12) is solved outward. Then, spacetime is assumed to be asymptotically Schwarzschild at the far boundary of the numerical domain, and α is set to $1/a$. Elsewhere, α is obtained by integrating Eq. (13) inward toward the origin. [Note that Eq. (14) is used only for convergence testing in this analysis].

This calculation is repeated at the end of each Runge-Kutta stage of each time step described in Secs. II B 2 and II B 3.

For supercritical solutions, these were the only two variables needed for determining the formation of a horizon and determining the mass of the resultant black hole. All that is interior to the horizon is marked by the collapse of the lapse in the chosen coordinates. The coordinate of the horizon is marked by a sharp spike in the variable a . Due to the spherical symmetry, the mass of the resultant black hole can be determined with the r coordinate of the maximum value of the variable a through a small rearrangement of the Schwarzschild radius relation to

$$M = \frac{r}{2}. \quad (15)$$

Notably, the collapse of the lapse for all points interior to the horizon causes the evolution to essentially freeze. This allows for continued stable evolution, without specialized coordinates, of the solution.

2. Discontinuous Galerkin solution

For building a DG version of massless scalar field collapse, the deal.II library [42] was chosen due to its being an open source library with an active community and a robust collection of tutorials. Built-in capabilities of deal.II used in this analysis were adaptive mesh refinement, dimension-agnostic mesh structure (one can take advantage of symmetries by using fewer dimensions), and matrix-free calculations. The matrix-free format calculates matrix elements on the fly instead of pulling from memory to increase computational speed.

The method of solving Eqs. (9) and (10) follows much of the traditional DG approach. First, the equations need to be written in conservative form. Explicitly, the right-hand side

of Eq. (9) is expanded out to a conservative form. Then, both sides are multiplied by a test function, v :

$$\int_{r_i}^{r_{i+1}} v \frac{\partial \Pi}{\partial t} dr = \int_{r_i}^{r_{i+1}} \left(v \frac{\partial}{\partial r} \left(\frac{\alpha}{a} \Psi \right) + 2v \frac{\alpha \Psi}{ra} \right) dr, \quad (16)$$

$$\int_{r_i}^{r_{i+1}} v \frac{\partial \Psi}{\partial t} dr = \int_{r_i}^{r_{i+1}} v \frac{\partial}{\partial r} \left(\frac{\alpha}{a} \Pi \right) dr. \quad (17)$$

Now the equations are integrated by parts to produce

$$\int_{r_i}^{r_{i+1}} v \frac{\partial \Pi}{\partial t} dr = \int_{r_i}^{r_{i+1}} \left(-\frac{\partial v}{\partial r} \frac{\alpha}{a} \Psi + 2v \frac{\alpha \Psi}{ra} \right) dr + v \frac{\alpha}{a} \Psi^* \Big|_{r_i}^{r_{i+1}}, \quad (18)$$

$$\int_{r_i}^{r_{i+1}} v \frac{\partial \Psi}{\partial t} dr = \int_{r_i}^{r_{i+1}} -\frac{\partial v}{\partial r} \frac{\alpha}{a} \Pi dr + v \frac{\alpha}{a} \Pi^* \Big|_{r_i}^{r_{i+1}}, \quad (19)$$

where Ψ^* and Π^* are the numerical flux. Since these equations form a linear system of hyperbolic equations, the Riemann conditions [27] must be solved to form the upwinding flux. This results in the following numerical fluxes, where an upper index $+$ indicates the value outside of the cell and an upper index of $-$ indicates the value within the cell, and $\hat{n}^+$ represents the outward normal of the element (either 1 or -1):

$$c^* = \frac{\alpha^+ \alpha^-}{a^+ \alpha^- + a^- \alpha^+}, \quad (20)$$

$$\Psi^* = c^* \left(\frac{a^+ \Psi^+}{\alpha^+} + \frac{a^- \Psi^-}{\alpha^-} + (\Pi^+ - \Pi^-) \hat{n}^+ \right), \quad (21)$$

$$\Pi^* = c^* \left(\frac{a^+ \Pi^+}{\alpha^+} + \frac{a^- \Pi^-}{\alpha^-} + (\Psi^+ - \Psi^-) \hat{n}^+ \right). \quad (22)$$

Ψ and Π are discretized as the sum of local discrete solutions of each element, k :

$$\Psi(x, t) \simeq \sum_{k=1}^K \Psi^k(x, t), \quad (23)$$

$$\Pi(x, t) \simeq \sum_{k=1}^K \Pi^k(x, t). \quad (24)$$

Ψ^k and Π^k are the solutions within each element. These solutions are represented by a collection of basis functions, ℓ , associated with different nodes within the element. The nodes are spaced out at Gauss-Lobatto nodes for increased stability, as is typical of the discontinuous Galerkin method. At each node, only one of the basis functions is nonzero. Each of these basis functions has a weight associated with it, $\hat{\Psi}_n^k$ and $\hat{\Pi}_n^k$, which can also be viewed as a nodal value

due to the association between nodes and basis functions. The basis functions used are Lagrange interpolation polynomials that are unity at a related node within a cell and zero at all other nodes within the cell:

$$\Psi^k(x, t) \approx \sum_{n=1}^{N_E} \hat{\Psi}_n^k(t) \ell_n(x), \quad (25)$$

$$\Pi^k(x, t) \approx \sum_{n=1}^{N_E} \hat{\Pi}_n^k(t) \ell_n(x). \quad (26)$$

Here, N_E is the total number of nodes within an element k and must be at least one larger than the polynomial order of the basis functions.

Incorporating the discretizations along with defining the test function v as one of the Lagrange interpolation polynomials gives

$$\int_{r_i}^{r_{i+1}} \ell_i \ell_j dr \frac{\partial \hat{\Pi}_j^k}{\partial t} = \int_{r_i}^{r_{i+1}} \left(-\frac{\partial \ell_i}{\partial r} \frac{\alpha}{a} \Psi^k + 2\ell_i \frac{\alpha \Psi^k}{ra} \right) dr + \ell_i \frac{\alpha}{a} \Psi^* \Big|_{r_i}^{r_{i+1}}, \quad (27)$$

$$\int_{r_i}^{r_{i+1}} \ell_i \ell_j dr \frac{\partial \hat{\Psi}_j^k}{\partial t} = \int_{r_i}^{r_{i+1}} -\frac{\partial \ell_i}{\partial r} \frac{\alpha}{a} \Pi^k dr + \ell_i \frac{\alpha}{a} \Pi^* \Big|_{r_i}^{r_{i+1}}, \quad (28)$$

where i and j represent iterations through all the basis functions within an element. They are so written to be suggestive of how the integrals can be rewritten as matrices and represent the two systems as

$$M_{ij} \frac{\partial u_j^k}{\partial t} = b_i, \quad (29)$$

where u_j^k can represent either Π_j^k or Ψ_j^k .

The mass matrix M_{ij} is block diagonal in DG, allowing for more efficient matrix inversion. However, to maximize the benefits of DG, a change of basis was used to make the mass matrix diagonal and therefore trivially inverted [43].

Now, both equations can be formed into an equation appropriate for explicit Runge-Kutta evolution. DG is constrained by the need for a strong stability-preserving Runge-Kutta method. For these simulations, a fourth-order, five-stage Runge-Kutta method was used [44]. One additional stage is needed when compared to the traditional Runge-Kutta fourth-order scheme; however, the formulation is built such that all stages do not need to be saved independently, low-storage Runge-Kutta, to further optimize the evaluation.

The time step size is uniform for the entire mesh and is determined by the smallest refinement level using the Courant-Friedrichs-Lewy condition [45].

For boundary conditions, the process for determining the solution to Eqs. (20)–(22) must be specified. Parity conditions are used on the inner boundary. Ψ is odd and all other variables are even. On the outer boundary, the solution was assumed to be continuous such that $\Psi^- = \Psi^+$, as well as for all other variables. No other condition was needed.

With the evolution of the dynamic variables set, the evolution of the adaptive mesh remains. At the end of every time step, a quick estimate of error was calculated for every cell in the mesh. If the error was greater than a user-prescribed tolerance, the time step was thrown out and all cells that had error greater than the prescribed value are flagged for refinement. In addition, any cell with error less than the user-specified value by a factor of 40 are flagged for coarsening. Then, the mesh was refined and coarsened following cell flags. The scalar field variables Ψ and Π were then moved from the old mesh to the new mesh with the embedded values for new refined cells and through a L^2 projection for coarsened cells. Next, the nondynamic equations were solved as prescribed in Sec. II B 1. Finally, the original time step is repeated with the more refined data. For the quick estimate of error in a cell, the L^2 norm of the left-hand side of Eq. (13) was used with a different quadrature rule that has an additional quadrature point.

Importantly, the integral for the L^2 norm of the left-hand side of Eq. (13) did take into account the physical structure of the simulation, meaning the spherical integrand term $r^2 dr$ was included here. This is different from all other integrals in this method and different from the quick estimate of error used in FD. The difference was needed as DG was unable to resolve the critical point without the change. This is believed to be because of the difference in dissipation between the two methods. FD employs an *ad hoc* solution of Kreiss-Oliger dissipation. This DG code employed no *ad hoc* dissipation. The dissipation came from the numerical flux, which was derived from the partial differential equations themselves. This may have allowed small errors from the outer region of the mesh to propagate inward, growing as they did due to the spherical symmetry. Changing the quick estimate of error to properly weigh regions in accordance with the underlying geometry resolved all issues.

3. Finite-differencing solution

For the fully adaptive FD code, the domain is subdivided into a binary tree of cells, each containing a uniform grid of nodal points storing the value of each field. The spatial derivatives in Eqs. (9) and (10) are computed via uniform fourth-order centered finite-difference stencils, interpolating values from neighboring cells as needed. This system is then integrated in time using the traditional fourth-order Runge-Kutta method, with fifth-order Kreiss-Oliger

dissipation to dampen out spurious numerical oscillations and ensure long-term stability.

Π is an even function and Ψ is an odd function, which is strongly enforced on the inner boundary via ghost nodes. For the outer boundary, the code uses asymmetric stencils to approximate spatial derivatives. The code also applies Sommerfeld boundary conditions to allow outgoing wavefronts to pass through the boundary with minimal reflection [46]:

$$\partial_t f = - \left[\partial_r f + \frac{f - f_0}{r} \right]. \quad (30)$$

Here Ψ_0 and Π_0 are both taken to be 0.

In order to perform adaptive mesh refinement, an interpolating wavelet-based refinement criterion inspired by Fernando *et al.* [47] is used. For a cell on a given level, each node either coincides with a node on the next-coarsest level or is the midpoint of two nodes on the next-coarsest level. The detail coefficients of a cell are defined to be the absolute difference between the value of a function at each midpoint and the value found by interpolating that function from points on the next-coarsest level. Every 20 time steps, the code computes these detail coefficients on each cell. If the maximum detail coefficient on a cell is below a coarsening threshold (10^{-10}) or above a refinement threshold (10^{-8}), the code flags that cell for coarsening or refinement, respectively. This refinement criterion focuses computational resources in regions with large spatial variation and bounds the error introduced by the FD stencil approximations.

III. RESULTS

A. Mass scaling

For all three families, the amplitude was varied in order to find a singular point, the critical point, separating black hole formation and dispersion of the scalar field [48]. Figures 1 and 2 show the mass scaling relations for all three different families using both FD and DG methods. All figures are on a logarithmic scale so that the critical exponent γ can be extracted as the slope. A best fit was performed for a simple straight line. The approximate error of slopes in Fig. 1 is 0.0003. The approximate error of slopes in Fig. 2 is 0.002. The errors were determined by fitting a straight line to the data, separating the data into regions of similar variance from the fit, setting the error in those regions to be the root mean square of the residuals, repeating the process three times, and using the variance of the final fit to determine the approximate error. More accurate fits are applied in Sec. III C that account for fine structure. Regardless of process, for both methods and all families, the slopes agree well with the accepted value of $\gamma \approx 0.37$ [49] and appear to be universal. Most notable is the difference in behavior between the two methods as the

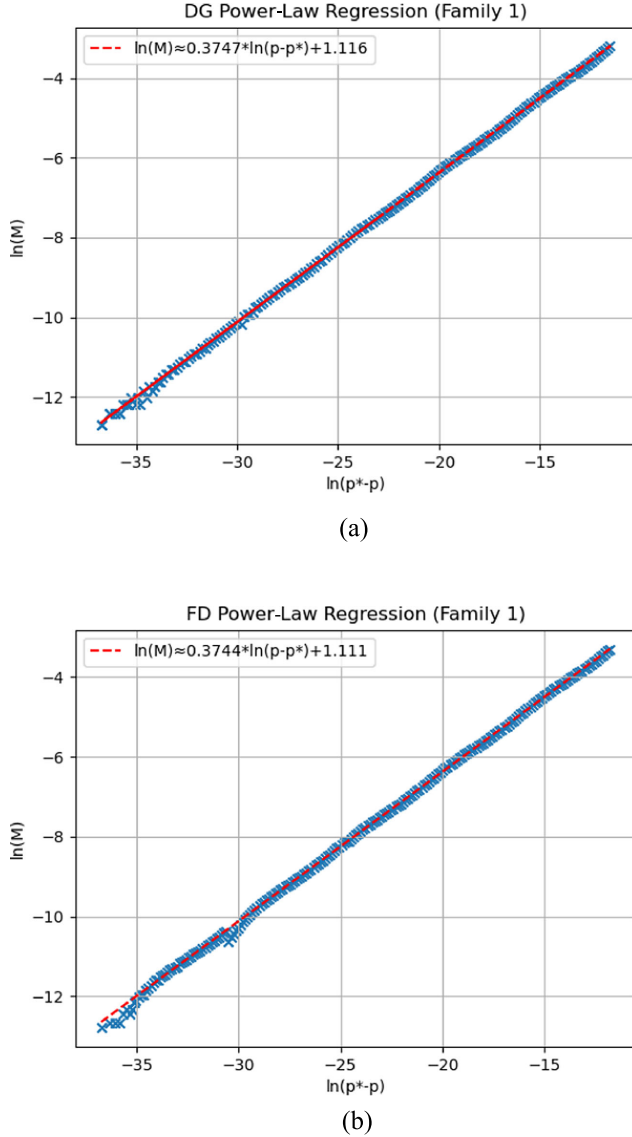


FIG. 1. Comparison of power-law scaling for family 1 from Table I. Up to 24 levels of refinement were allowed for these simulations, which was the most performed for any run in this paper. Both slopes carry an error of approximately ± 0.0003 . (a) Power-law scaling of the black hole mass forming from a massless scalar field of family 1 collapsing with 24 allowed refinement levels using DG. (b) Power-law scaling of the black hole mass forming from a massless scalar field of family 1 collapsing using a full adaptive refinement FD.

adaptive meshes approach their limits. This is seen in the points closest to criticality. FD results have a “droop” that occurs twice in the FD graph of Fig. 1 and once at the far left of both FD graphs in Fig. 2. DG results in these same regions also experience some instability. However, the instability appears more like randomized noise instead of falling off the trend line before abruptly rejoining. This appears to be caused by the difference in dissipation or different refinement weights as explained in Sec. II B 2.

These results included three different families of initial data. All three of these families resulted in critical exponents compatible with those of Choptuik’s original work, which also included four of its own families. This shows the universality of the collapse, independent of initial data, of the spherically symmetric massless scalar field.

B. Discrete self-similarity

Figure 3 shows the value of the scalar field Φ plotted with respect to the negative of logarithmic time τ . (The negative is so that, more naturally, time increases going right.) The amplitude was chosen to be the largest amplitude of a dispersing field for the respective methods. As can be seen, this produces a figure with periodic behavior. The wavelength of this behavior is the discrete spacing of self-similarity, Δ , also called the echoing constant [49].

To determine the error associated with the echoing constant, the wavelength was measured at three different points. The reported values are the average and standard deviation of those measurements. The wavelength measurement was performed at zero crossings, avoiding the first and final two of the evolution, as strong-field dynamics did not dominate at these moments. A linear approximation was used to find the approximate coordinates of the zero crossing. For DG, $\Delta = 3.451 \pm 0.007$. This is in line with the expected value of $\Delta \simeq 3.44$ [49]. FD methods also agree with a measured value of $\Delta = 3.451 \pm 0.007$. No significant difference has been noted between the two methods.

C. Fine structure

The same data as plotted in Fig. 1 were fit to a model of $y = mx + b + A \cos(\omega x + p)$ for both methods. Data farther away from the critical point were used to minimize numerical noise. The linear terms were removed both from the data and the best-fit line to produce Fig. 4.

From the best fit of the DG data, values for the critical exponent and fine structure wavelength were extracted, $\gamma = 0.37408 \pm 0.00002$ and $\varpi = 4.6283 \pm 0.0039$. An additional value for ϖ can be calculated using the stated relation $\varpi = \Delta/(2\gamma)$ using $\Delta = 3.451 \pm 0.007$ from Sec. III B as $\varpi = 4.6126 \pm 0.0094$. Both values for ϖ agree with each other and the expected value of $\varpi \approx 4.6$ [16,17].

Repeating the same process for the FD data leads to measurements of $\gamma = 0.37408 \pm 0.00002$ and $\varpi = 4.62222 \pm 0.0029$. Again, this measured value of ϖ agrees with the calculated value using the relation between Δ and γ , $\varpi = 4.6126 \pm 0.0094$. Again, no serious difference is noticed between the two methods in the results.

D. Computation

A comparison of convergence rates is shown in Fig. 5. The figure shows the maximum error on the mesh at the same central observer time for simulations limited by a maximum refinement ranging from 9 to 16. Then, this

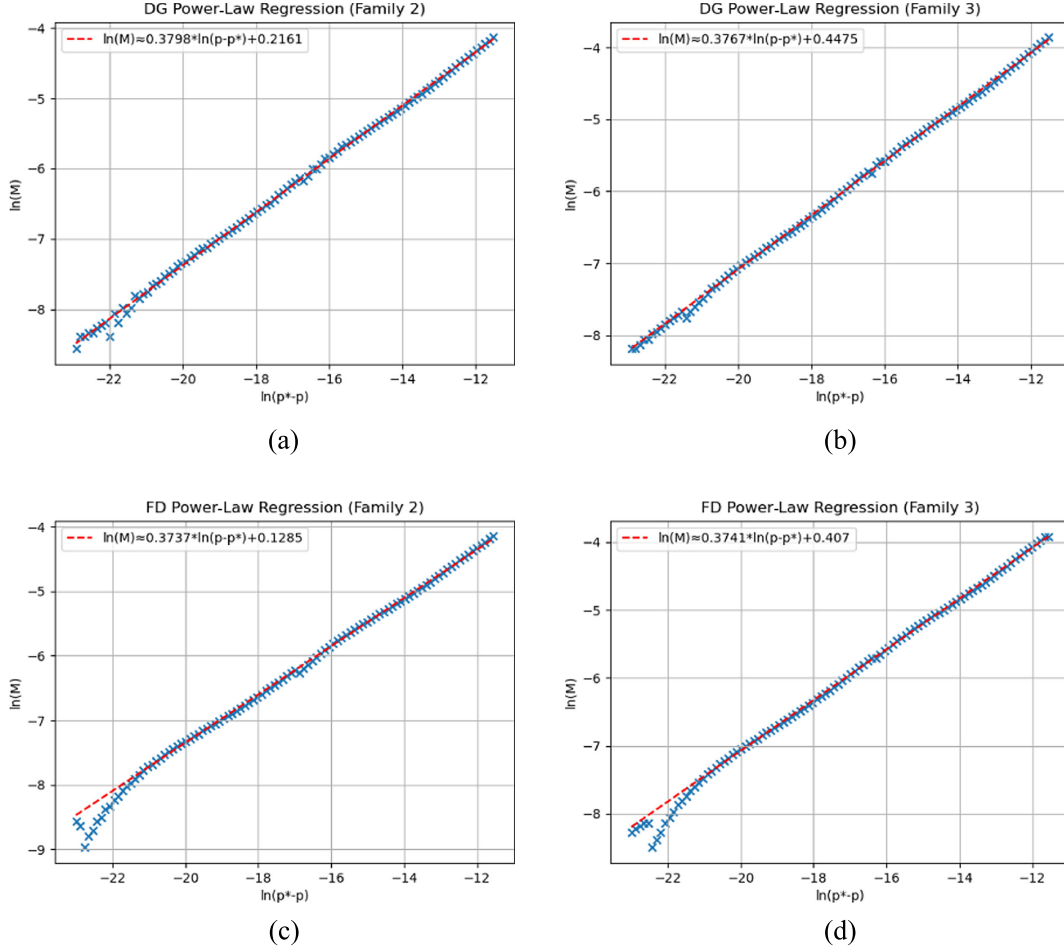


FIG. 2. Collection of power-law scaling for families 2 and 3 from Table I. Columns correspond with the families, and rows correspond with the codes used to run the simulations. The first row used the DG code; the second used the new adaptive FD code. These were run with a lower allowed max refinement of 19. All slopes carry an approximate error of ± 0.002 . (a) Power-law scaling of the black hole mass forming from a massless scalar field of family 2 collapsing with 19 allowed refinement levels using DG. (b) Power-law scaling of the black hole mass forming from a massless scalar field of family 3 collapsing with 19 allowed refinement levels using DG. (c) Power-law scaling of the black hole mass forming from a massless scalar field of family 2 collapsing using a full adaptive refinement FD. (d) Power-law scaling of the black hole mass forming from a massless scalar field of family 3 collapsing using a full adaptive refinement FD.

value was plotted against the number of degrees of freedom at that moment. Error was measured as the violation of Eq. (14) at each individual point. The central observer time, the proper time of an observer at $r = 0$, at the end of each run was 6.4, where $T_0^* \approx 6.547$, the earliest time of black hole formation, for the family used. Since the methods were both built to be fourth-order accurate and employed adaptive refinement, their respective slopes should be less than -4 on the logarithmically scaled graph. The simulations were performed using family 1 from Table I and an amplitude of 0.33603529, the critical point out to eight significant figures. The number of significant figures matches the threshold used for refining cells, 10^{-8} . (Both methods were tuned to a critical point of 16 digits; only the first eight were used above, as the threshold would imply these are the digits we have confidence in. DG had a

critical amplitude of 0.3360352960145161. FD had a critical amplitude of 0.3360352960560983.) This also explains the end behavior of both lines. The pointwise error begins to approach the threshold, and numerical noise then enters and causes the maximum error to oscillate near the limit.

Both methods have a slope less than -4 and are approximately equal. The largest difference is the number of degrees of freedom needed between the two methods. The difference in number of degrees of freedom is not representative of the comparison of the two methods since the runs stop short of T_0^* . The difference represents the initial overhead of DG. This initial difference is overcome as more accuracy is required, as will be shown in the ensuing figures. The cause of the initial overhead of DG is twofold. First, DG double counts the degrees of freedom

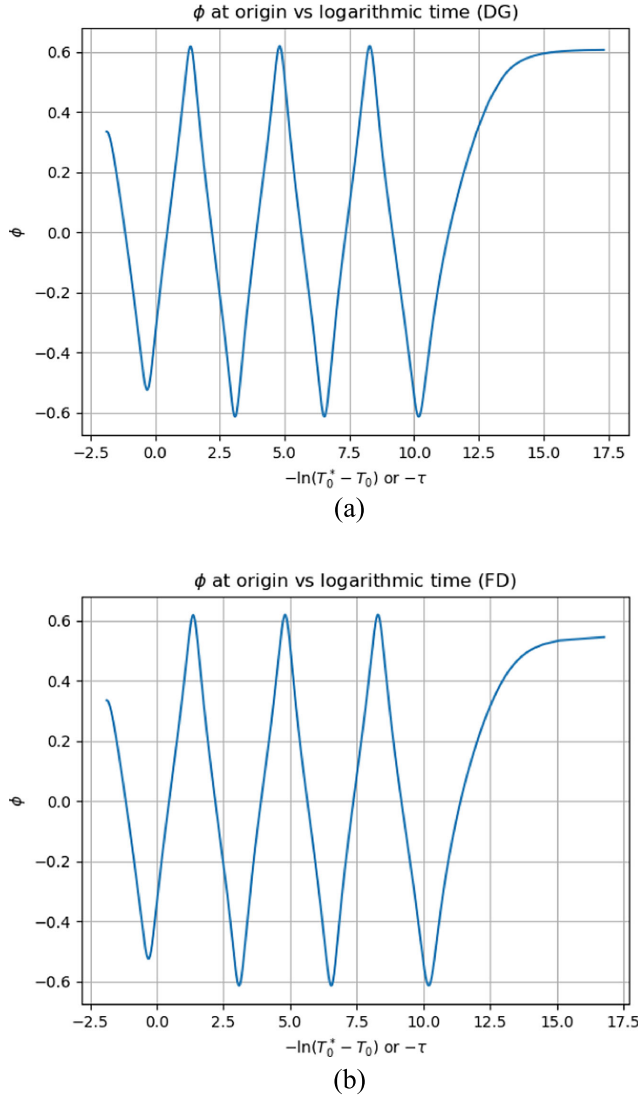


FIG. 3. As strong-field dynamics take hold, echoing can be seen as the formation of periodic behavior that then tapers off as the subcritical field disperses. (a) The value of the scalar field, Φ , at the origin as seen by a central observer for the largest amplitude subcritical DG run. (b) The value of the scalar field, Φ , at the origin as seen by a central observer for the largest amplitude subcritical FD run.

between cells to allow discontinuity. Second, the refinement process differs between the two methods as explained in Sec. II B 2. In particular, the DG refinement method assigned higher weights to points farther away from the origin than the FD method. Extra refinement in outer regions is caused by this difference, leading to more degrees of freedom being used.

The potential of DG is first seen in Fig. 6. This graph represents a single run in both methods using family 1 with the largest amplitude dispersing wave, where the number of degrees of freedom is plotted against the central observer time. Degrees of freedom are used as a heuristic for the

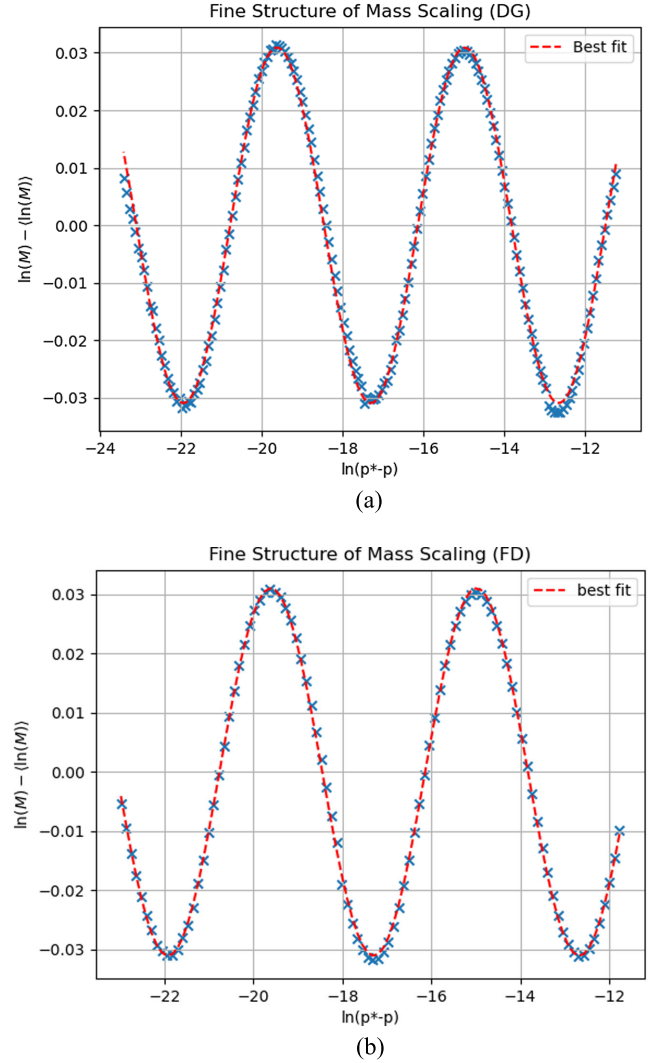


FIG. 4. The figure shows the periodic behavior of the fine structure of the mass scaling for both methods. For the y axis, $\langle \ln(M) \rangle$ represents the linear terms of the best fit, following the same format as Hod and Piran [16]. (a) Image of the fine structure of the mass scaling seen in the DG simulations. (b) Image of the fine structure of the mass scaling seen in the FD simulations.

necessary computational power since this correlates with the amount of storage and the number of computations that need to be performed. DG initially uses more degrees of freedom but is surpassed by FD as the simulation approaches the critical moment deciding between dispersion and collapse. The maximum number of degrees of freedom used by FD is 32% higher than that used by DG. As shown by the inset in Fig. 6, the true peak appears slightly after the accumulation time of a supercritical solution, T_0^* . The true peak should not be disregarded, as in a critical search, T_0^* is not known in advance, and the true peak still appears in the immediate neighborhood of T_0^* .

Figure 7 provides a better comparison between methods with relation to computational time. The figure again

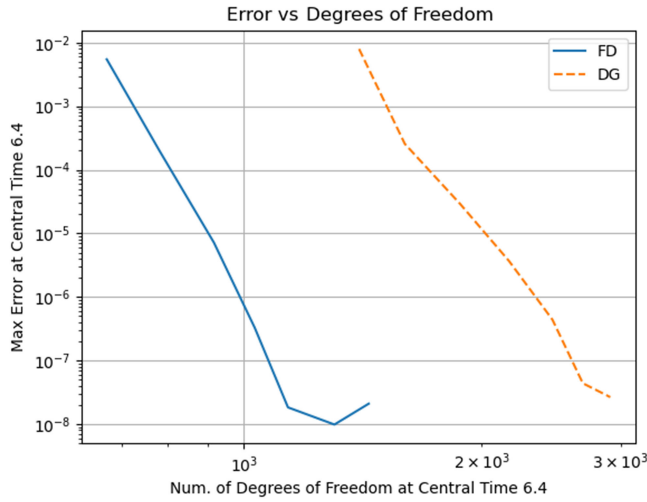


FIG. 5. The L^∞ norm of the right-hand side of Eq. (14) of FD and DG methods run at maximum refinements allowed ranging from 9 to 16. The simulations were run until central observer time of 6.4, less than $T_0^* \approx 6.547$ for the family.

represents a single run in both methods using family 1 with the largest amplitude dispersing wave. However, degrees of freedom are now plotted against the temporal step number. The temporal step number more accurately aligns with the proportion of the simulation expended at the corresponding degree of freedom. Both simulations were run to a central observer time of 6.55, just past the accumulation time for family 1, $T_0^* \approx 6.547$. This is done to capture the most computationally demanding section of the simulation. Strikingly, the DG simulation used 49% fewer steps than FD. Compounding upon the advantage of fewer steps, DG also used fewer degrees of freedom on average. The mean

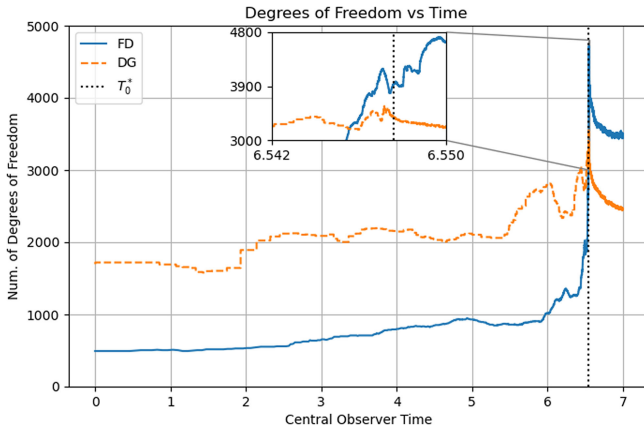


FIG. 6. The total number of degrees of freedom used in the FD code and the DG code versus the time as seen by a central observer of the simulation using family 1 and the amplitude of the largest dispersing wave of each method. The accumulation time T_0^* is indicated with a vertical line. This shows the initial overhead of DG and the ability of DG to overcome this overhead as more accuracy is needed.

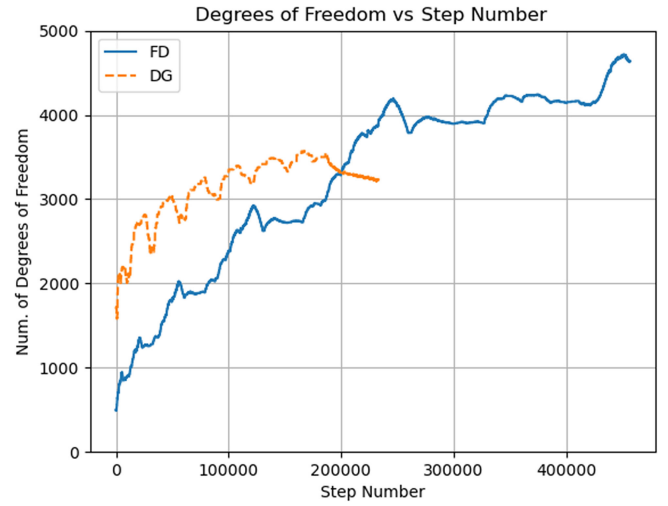


FIG. 7. The total number of degrees of freedom used in the FD code and the DG code versus the time step number of the simulation, using family 1 and an amplitude of the largest dispersing wave. Both simulations were run to a central observer time of 6.55 with $T_0^* \approx 6.547$ for this family.

number of degrees of freedom per step for DG was 3134.4, whereas FD used 3256.6 degrees of freedom on average. This suggests that not only does DG take significantly fewer steps, but they are slightly less computationally intensive steps as well.

Figure 8 aids in demonstrating where the large difference in step numbers comes from. This figure again uses the same runs from the previous two figures. Now, the refinement level is contrasted with the step number. Increasing the level of refinement will decrease the smallest size of a

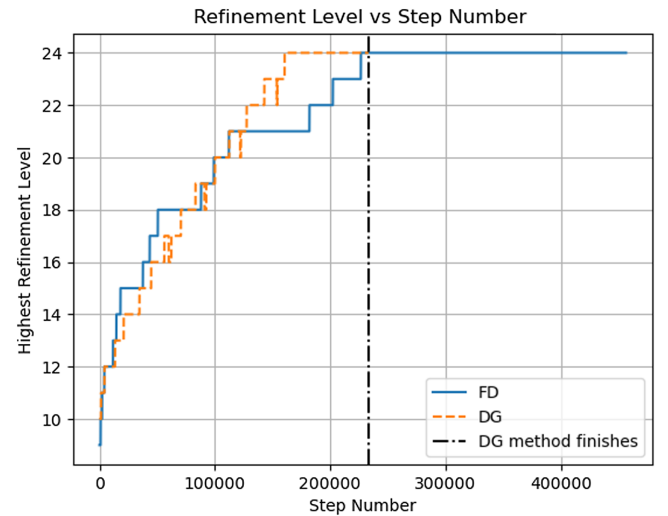


FIG. 8. The highest level of refinement used in the FD code and the DG code versus the time step number of the simulation, using family 1 and an amplitude of the largest dispersing wave. Both simulations were run to a central observer time of 6.55 with $T_0^* \approx 6.547$ for this family.

cell by a factor of 2. Since the Courant-Friedrichs-Lewy condition [45] demands the temporal step be smaller than the length of the smallest cell, a larger refinement level produces a slower moving simulation. Figure 8 clearly shows that the DG simulation, proportionally, spends more of the simulation at lower refinement levels. The difference in necessary refinement levels signals the ability of DG to resolve complex solutions with fewer degrees of freedom. The benefits would grow with higher demands for accuracy and an increase in the complexity of simulated fields.

Ten simulations were run for both the DG and FD codes. They were run on the same machine, with a single core, and for the largest dispersing wave. The average and standard deviations were calculated from the ten runs for the reported run times. The DG method took 1057.946 ± 24.061 s. The FD method took 3176.893 ± 13.908 s. The DG code ran over 3 times as fast.

The results suggest that the benefits of DG will grow for critical phenomena simulations in higher dimensions and of more complicated fields. Such simulations are frequently limited by precision and refinement levels, which is where DG excels.

IV. CONCLUSION

Critical phenomena in gravitational collapse provide key tests of Einstein's field equations in the strong-field regime. As investigations expand into more general systems, computational resources must be optimized to fully understand the universality of critical behavior. To test the viability of the DG method in critical phenomena of gravitational collapse, the classic example of massless scalar field collapse was performed and compared to results

using the FD method. DG was able to replicate the same results as the more traditional FD approach for the power-law scaling relation, the discrete self-similarity constant, and the period of fine structure of the power-law scaling. DG does have a larger initial computational cost; however, it tends to refine less and requires fewer degrees of freedom, both on average and at maximum, when compared to FD, leading to more efficient and accurate analyses of critical phenomena. DG excels the closer the simulation is to the critical point, where the number of temporal steps used is almost halved and run time is 3 times as fast. This, compounded with the ability to flag for the adaptive mesh refinement more frequently, will directly aid in helping overcome run time and resolution limitations characteristic of FD for more general systems. A detailed extension of nonlinear, axisymmetric gravitational wave collapse using DG will be reported in future work.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [48].

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